Fulton Science Math Tournament

October 2026 - Team Round

Honor Pledge

As a participant of FSMT, I promise to act with honesty, integrity, and respect in every challenge and competition I face. I will not seek or offer unauthorized assistance, and I will ensure that all work I submit is truly my own. I understand that fairness and trust are essential to the spirit of mathematics, and I accept full responsibility for upholding these values. I acknowledge that any breach of this pledge may lead to disqualification, and I am committed to competing with honor, diligence, and respect for my fellow competitors.

Team Captain's Name _____ Date _____

School _____

Team Name _____

PLEASE DO NOT BEGIN UNTIL BEING INSTRUCTED TO DO SO

The Team Round is an intensive, collaborative session designed to challenge a group of four students with a series of diverse mathematical and logical problems. During a 20-minute period, the team must work together to solve 10 unrelated problems, which span a wide range of topics and difficulty levels. Questions are aimed to challenge students and require clever thinking and creative ideas in order to solve.

It is important to note that in every written round of the competition, the answer must include the unit specified in the answer blank. These units are always presented in their plural form, regardless of whether a singular form might seem more appropriate based on the context of the problem. Answers that fail to include the exact unit as indicated will not be accepted. For additional clarification and a comprehensive explanation of rules, procedures, and scoring, participants should consult the Rules and Format Page provided. If you are still uncertain about a question, an answer, or the required format, ask a proctor for clarification immediately.

Team Score _____ **Scorer's Initials** _____

1. _____ Alex and Joshua each have a number. The sum of their numbers is 18, and their product is 45. If Alex's number is smaller than Joshua's number, what are their numbers? Express your answer as an ordered pair (Alex's number, Joshua's number).

2. _____ points Eric Jung plays League of Legends for 4 hours each weekday. He earns 30 points for every hour he plays. However, he loses 40 points each Saturday and each Sunday. Assuming there are exactly 52 weeks in a year, how many net points does he gain in one year?

3. _____ handshakes There are 19 people at a conference: 13 men and 6 women. Each man shakes hands with every woman and with every other man exactly once. The women do not shake hands with each other. How many handshakes occur in total?

4. _____ eggs A turtle can lay either 1, 2, 3, or 4 eggs every time it lays eggs, with equal probability for each. If a turtle lays eggs 3 times, what is the expected total number of eggs? Express your answer as a decimal.

5. _____ diagonals How many diagonals does a 17-gon have? An edge does not count as a diagonal.

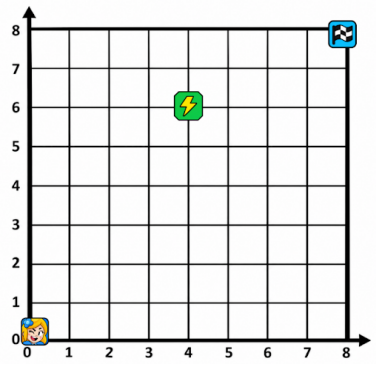
6. _____ times Starting at $12:01$, how many times will the hour and minute hands of a clock overlap again during the next 12 hours?

7. _____ seconds Ivane and Aran are running in opposite directions on a 500-meter circular track. Aran runs at 4 meters per second, and Ivane runs at 3.2 meters per second. Both start from the same point, but Ivane starts running before Aran. How many seconds earlier must Ivane start so that they meet for the first time when Aran has run 250 meters? Express your answer as a common fraction.

8. _____ points A dartboard consists of 3 concentric circles with radii 2, 4, and 7, forming three scoring regions. The point values of the three regions are the smallest positive integers proportional to the regions' areas. Bob may throw as many darts as he wants, and every dart lands in one of the scoring regions. Bob's opponent scored 94 points. What is the minimum number of points Bob can score while still beating his opponent?

9. _____ ways

Brawl Stars has a map maker. Sahil made a map that is 8 blocks by 8 blocks. Athena Piper starts at $(0, 0)$ and wants to reach $(8, 8)$. A power cube is located at $(4, 6)$. She can only move one unit up or one unit right at a time, so she always moves between integer-coordinate points. In how many ways can she reach $(8, 8)$ while also collecting the power cube?



10. _____

A positive integer n has exactly 12 positive divisors. What is the maximum possible number of positive divisors of n^2?